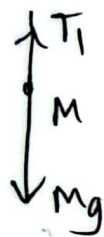
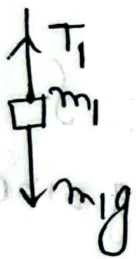


Given that,

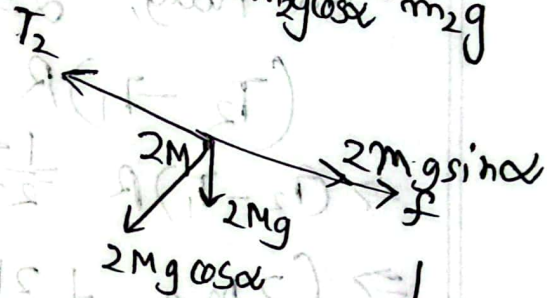
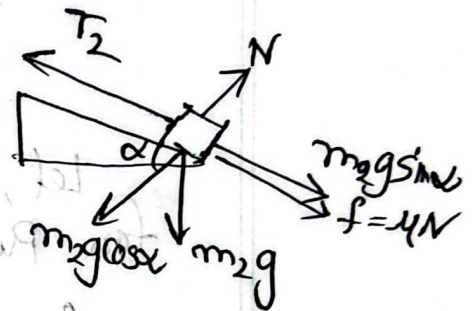
$R = 50 \text{ cm} = 0.5 \text{ m}$, uniform mass $M = 1 \text{ kg}$ (disk)

a) $\Sigma \tau = I \alpha = 0$

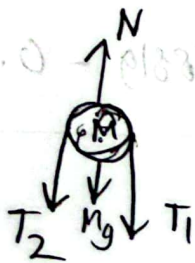
block M (vertically hanging)



Block $2M$ (on incline)

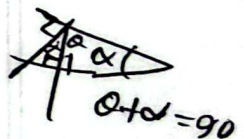


pulling (disk)



$$\tau = (T_2 - T_1) R$$

$$I = \frac{1}{2} M R^2 \text{ disk}$$



b.

for M block,

$$Mg - T_1 = Ma \quad (1)$$

$$\Rightarrow T_1 = Mg - Ma$$

for 2M block,

$$T_2 - 2Mg \sin \alpha - \mu 2Mg \cos \alpha = 2Ma$$

$$\Rightarrow T_2 = 2Ma + 2Mg \sin \alpha + \mu 2Mg \cos \alpha$$

Let, angular acceleration α for Pulley, $\alpha = \frac{a}{R}$

$$(T_2 - T_1)R = I\alpha$$

$$\Rightarrow (T_2 - T_1)R = \frac{1}{2}MR^2 \cdot \frac{a}{R}$$

$$\Rightarrow (2Ma + 2Mg \sin \alpha + \mu 2Mg \cos \alpha) - (Mg - Ma) = \frac{1}{2}Ma$$

$$\Rightarrow a = \frac{Mg - 2Mg \sin \alpha - \mu 2Mg \cos \alpha}{M + 2M + \frac{1}{2}M}$$

$$= \frac{1 \cdot 9.8 - 2 \cdot 9.8 \cdot 0.258819 - 0.15 \cdot 2 \cdot 9.8 \cdot 0.9}{1 + 2 + 0.5}$$

$$= 0.53924 \text{ m/s}^2$$

$$\therefore \alpha = \frac{a}{R} = \frac{0.53924}{0.56} \text{ rad/s}^2$$

$$= 1.08 \text{ rad/s}^2$$

Date:/...../.....

c. from b, $a = 0.53924 \text{ m/s}^2$

$$T_1 = Mg - Ma = 9.26076 \text{ N}$$

$$T_2 = 2Ma + 2Mg \sin \alpha + \mu 2Mg \cos \alpha = 8.9915 \text{ N}$$

as, $T_1 > T_2$, T_1 is higher, so, $T = (T_1 - T_2) R$
if we consider direction opposite of a

$$T_{\text{net}} = (T_1 - T_2) R$$

$$= (9.26076 - 8.9915) \times 0.5 \text{ Nm}$$

$$= 0.13481 \text{ Nm}$$